

Build 4 APIs in basic Node.js using Express and MongoDB. You may use Google or AI tools to generate code, but you must understand every line: route, controller logic, request body, params, MongoDB query, validation, and response.

Expected for each API:

- Correct HTTP method and route
- Proper request body / params handling
- MongoDB create/read/update/delete usage
- Clean JSON response
- Basic error handling
- Short explanation of how the API works

Rithi – Student Management

1. POST /api/students – Create student
2. GET /api/students – List all students
3. GET /api/students/:id – Get student by ID
4. PATCH /api/students/:id/status – Update student status

Lokesh – Course Management

1. POST /api/courses – Create course
2. GET /api/courses – List all courses
3. GET /api/courses/category/:category – Filter courses by category
4. PATCH /api/courses/:id – Update course details

Princy – Task Tracker

1. POST /api/tasks – Create task
2. GET /api/tasks – List all tasks
3. GET /api/tasks/status/:status – Filter tasks by status
4. PUT /api/tasks/:id/complete – Mark task as completed

Gerauld – Employee Directory

1. POST /api/employees – Create employee
2. GET /api/employees – List all employees
3. GET /api/employees/department/:department – Get employees by department
4. PATCH /api/employees/:id/role – Update employee role

Karunya – Book Library

1. POST /api/books – Add book

2. GET /api/books – List all books
3. GET /api/books/author/:authorName – Get books by author
4. PUT /api/books/:id/issue – Mark book as issued

Vignesh – Product Inventory

1. POST /api/products – Add product
2. GET /api/products – List all products
3. GET /api/products/low-stock – Get products with low stock
4. PATCH /api/products/:id/stock – Update product stock

Kavin – Event Registration

1. POST /api/events – Create event
2. GET /api/events – List all events
3. GET /api/events/:eventId/registrations – Get registrations for an event
4. POST /api/events/:eventId/register – Register a participant for event

Submission:

Each intern should submit:

1. Source code
2. MongoDB collection structure
3. Sample request and response for all 4 APIs
4. 5-line explanation for each API
5. One screenshot from Postman or Thunder Client for each API